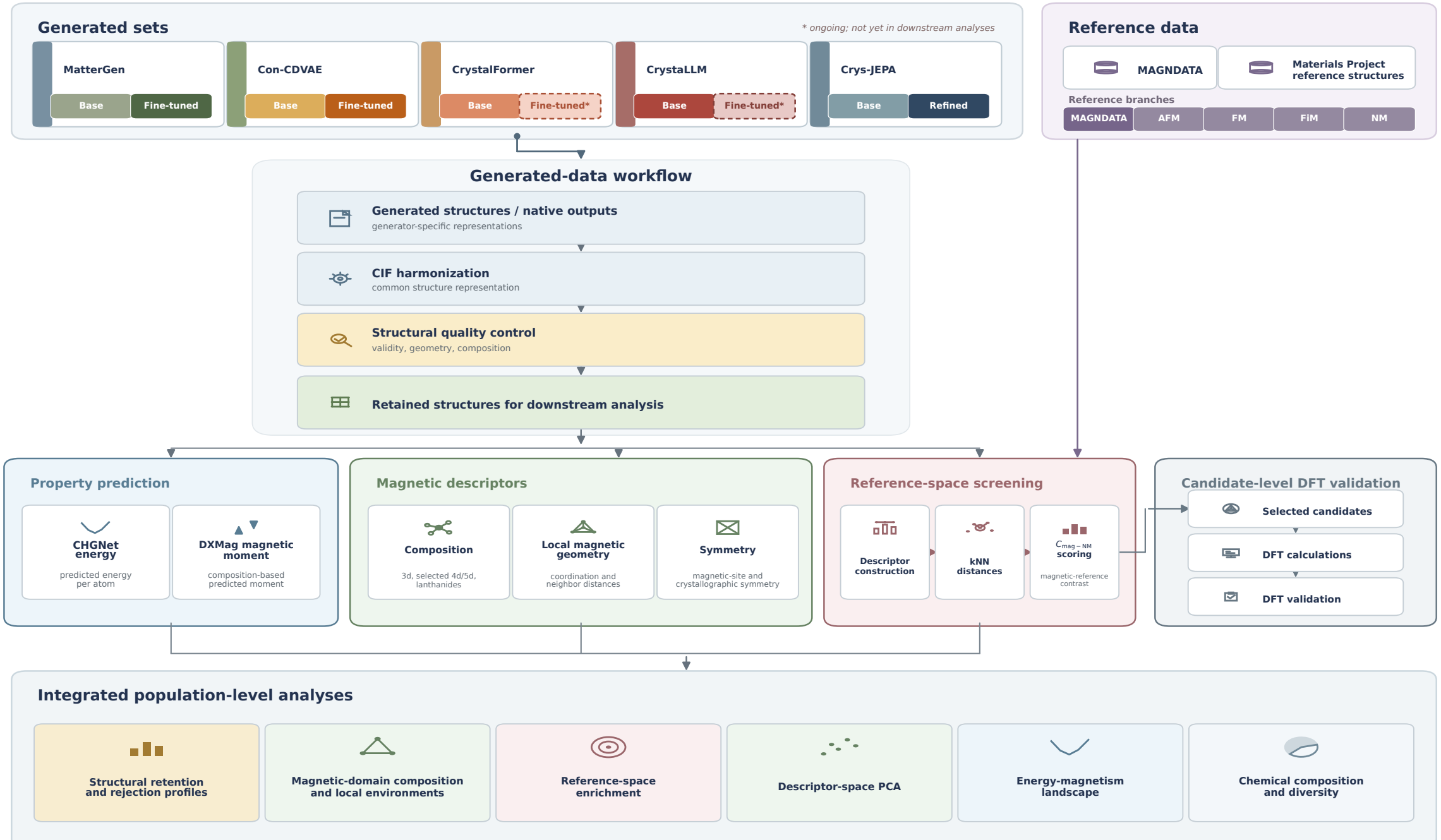

Generated sets
* ongoing; not yet in downstream analyses
MatterGen
Base
Fine-tuned
Con-CDVAE
Base
Fine-tuned
CrystalFormer
Base
Fine-tuned*
CrystaLLM
Base
Fine-tuned*
Crys-JEPA
Base
Refined
Reference data
MAGNDATA
Materials Project reference structures
Reference branches
MAGNDATA
AFM
FM
FiM
NM
Generated-data workflow
Generated structures / native outputs
generator-specific representations
CIF harmonization
common structure representation
Structural quality control
validity, geometry, composition
Retained structures for downstream analysis
Property prediction
CHGNet energy
predicted energy per atom
DXMag magnetic moment
composition-based predicted moment
Magnetic descriptors
Composition
3d, selected 4d/5d, lanthanides
Local magnetic geometry
coordination and neighbor distances
Symmetry
magnetic-site and crystallographic symmetry
Reference-space screening
Descriptor construction
kNN distances
C_{mag-NM} scoring
magnetic-reference contrast
Candidate-level DFT validation
Selected candidates
DFT calculations
DFT validation
Integrated population-level analyses
Structural retention and rejection profiles
Magnetic-domain composition and local environments
Reference-space enrichment
Descriptor-space PCA
Energy-magnetism landscape
Chemical composition and diversity